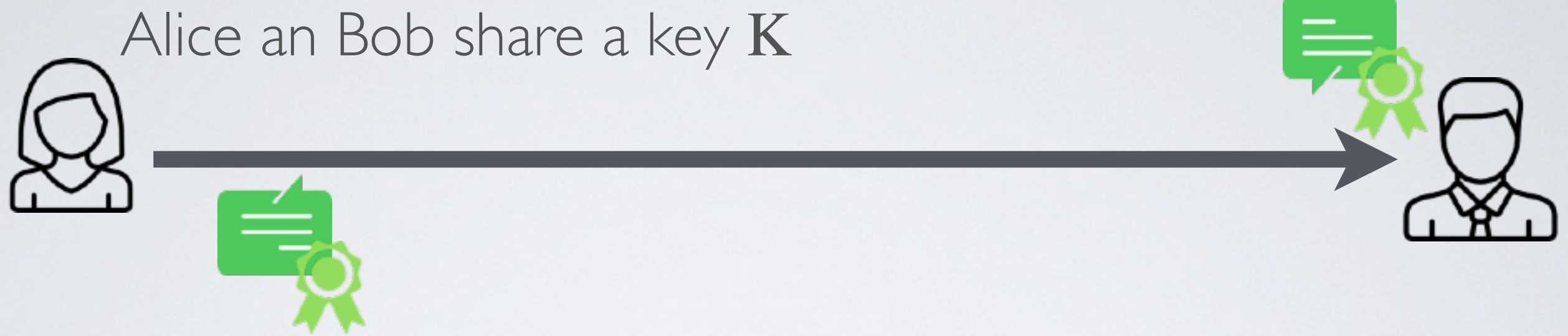


Authenticated Encryption (2013)



Encrypt-and-MAC (E&M)
considered vulnerable

$$AE_K(m) = E_K(m) \parallel H_K(m)$$

Old SSH

~~MAC then Encrypt (MtE)~~
considered vulnerable

$$AE_K(m) = E_K(m \parallel H_K(m))$$

TLS 1.0-1.1

Encrypt-then-MAC (EtM)
considered standard

$$AE_K(m) = E_K(m) \parallel H_K(E_K(m))$$

*AES-GCM
New SSH
TLS 1.2-1.3*

Digital Signatures